UNIT IV

17. (a) Show that

$(5x^4 + 3x^2y^2 - 2xy^3)\,dx + (2x^3y - 3x^2y^2 - 5y^4)\,dy = 0$.

(b) Solve $(a^2 - 2xy - y^2)\,dx - (x+y)^2\,dy = 0$.

Or

18. (a) Solve $y = (x-a)\,p - p^2$.

(b) Solve $y = px + \dfrac{a}{p}$.

UNIT V

19. Solve $(D^3 + D^2 + D + 1)\,y = 2x^3 + 3x^2 - 4x + 5$.

Or

20. Solve $(D^2 - 4D + 3)\,y = \sin 3x \cos 2x$.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018.

First Semester

MATHEMATICS – I

(Common to All Branches)

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Find the radius of curvature at any point $(at^2, 2at)$ of the parabola $y^2 = 4ax$.

2. Define the envolute and Involute.

3. Find the first partial derivatives of $z = x^3 + y^3 - 3axy$.

4. If u, v are functions of r, s and r, s are functions of x, y then prove that $\dfrac{\partial(u,v)}{\partial(x,y)} = \dfrac{\partial(u,n)}{\partial(r,s)} + \dfrac{\partial(r,s)}{\partial(x,y)}$.

5. Change the order of integration in $I = \int_{-a}^{a}\int_{0}^{\sqrt{a^2-y^2}} f(x,y)\,dx\,dy$.

6. Evaluate $\int_0^1\int_0^2\int_1^2 x^2 yz\,dx\,dy\,dz$.

7. Write down any rule for finding integrating factors.

8. Find the I.F of the equation $(1-x^2)\frac{dy}{dx}-xy=1$.

9. Solve $(D^3-6D^2+11D-6)Y=0$.

10. Define the homogeneous linear equation.

SECTION B — $(5 \times 11 = 55$ marks)

Answer ALL choosing ONE full question from each Unit.

UNIT I

11. Show that the radius of curvature at any point of the cycloid $x=a(\theta-\sin\theta), y=a(1-\cos\theta)$ is $4a\cos\frac{\theta}{2}$.

Or

12. Prove that $\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

UNIT II

13. If $V=(x^2+y^2+z^2)^{\frac{-1}{2}}$, then prove that $\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}=0$.

Or

14. Find the volume of the greatest rectangular Parallelopiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

UNIT III

15. (a) Find the area of a plate in the form of a quadrant of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=0$.

(b) Show that the area between the parabolas $y^2-=4ax$ and $x^2=4ay$ is $\frac{16}{3}a^2$.

Or

16. Find the volume of the portion of the sphere $x^2+y^2+z^2=a^2$ lying inside the cylinder $x^2+y^2=ax$.

UNIT IV

17. Solve $p^2 - p(\cos x + \sec x) + 1 = 0$. (11)

Or

18. Solve $x(1-x^2)\frac{dy}{dx} + (2x^3 - 1)y = x^3$. (11)

UNIT V

19. Solve by method of variation of parameter $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$. (11)

Or

20. Solve $\frac{dx}{dt} + y = \sin t$; $\frac{dy}{dt} + x = \cos t$ when $x = 2$, $y = 0$ at $t = 0$. (11)

5781001

B.Tech. DEGREE EXAMINATION,

NOVEMBER/DECEMBER 2017.

First Semester

MATHEMATICS — I

(Common for All branches)

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Write the Cartesian formula for radius of curvature.

2. Define gamma and beta function.

3. If $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ then find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

4. If $x = r\cos\theta$, $y = r\sin\theta$ then find $\frac{\partial(x,y)}{\partial(r,\theta)}$.

5. Evaluate $\int_0^1\int_0^2 (x^2 + y^2)\,dy\,dx$.

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6. Evaluate $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-y^2}} xdxdy$ by change of order of integration.

7. Solve $p = \tan(y - xp)$.

8. Solve $(px - y)(py + x) = 2p$.

9. Solve $(D^2 - 4D + 3)y = 0$.

10. Find the particular integral of $(D^2 + 1)y = \sin x$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full questions from each Unit.

UNIT I

11. (a) Find the radius of curvature at the point $\left(\frac{a}{4}, \frac{a}{4}\right)$ to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$. (5)

(b) Prove that the radius of curvature at the point $(a\cos^3\theta, a\sin^3\theta)$ on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is $3a\sin\theta\cos\theta$. (6)

Or

12. Prove that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} \times \int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta = \pi$. (11)

UNIT II

13. (a) If z is a function of x, y and u, v are the other two variable, such that $u = lx + my$, $v = ly - mx$. Show that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right).$$ (6)

(b) Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. (5)

Or

14. (a) A rectangular box open at the top, is to have a volume of 32cc. Find the dimensions of the box, the requires the least material for its construction. (6)

(b) Expand $e^x \log(1 + y)$ in powers of x and y up to terms of third degree. (5)

UNIT III

15. Change the order of integration in $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2 + y^2}}\, dydx$ and hence evaluate. (11)

Or

16. Find the volume of the region bounded by the surfaces $y^2 = 4ax$, $x^2 = 4ay$ and the plane $z = 0$, $z = 3$. (11)

B.Tech. DEGREE EXAMINATION, MAY 2017.
First Semester
MATHEMATICS – I
(Common to All branches)
(From 2013-14 onwords)

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Define curvature and radius of curvature.

2. Prove that $B(m,n) = B(n,m)$.

3. If $x^y + y^x = a^k$ find $\frac{dy}{dx}$.

4. If $u = 2xy, v = x^2 - y^2$ while $x = r\cos\theta, y = r\sin\theta$ prove that $\frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$.

5. Evaluate $\int\limits_{-1}^{2} \int_{x}^{x+2} dy\ dx$.

6. Change of the order of the integration $\int\limits_{0}^{\infty}\int\limits_{0}^{\infty} \frac{e^{-y}}{y} dy\ dx$.

7. Find the general solution of $y = xp + \frac{\alpha}{p}$.

8. Define exact differential equation.

9. Explain second order linear differential equation.

10. Solve $(D^2 - 3D + 2)y = e^{5x}$.

PART B – (5 × 11 = 55 marks)
Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. (a) Show that $\int_0^1 \frac{x^2dx}{\sqrt{1-x^4}} \times \int_0^\infty \frac{dx}{\sqrt{1+x^4}} = \frac{\pi\sqrt{2}}{4}$. (5)

(b) Find the centre of curvature and circle of curvature at $\left(\frac{\alpha}{4}, \frac{\alpha}{4}\right)$ on $\sqrt{x} + \sqrt{y} = \sqrt{\alpha}$. (6)

Or

12. (a) Obtain the equation of the evolute of the ellipse. (8)

(b) Prove that the radius of curvature at any point of the cycloid

$x = \alpha(\theta + \sin\theta), y = \alpha(1 - \cos\theta)$ is $4\alpha\cos\frac{\theta}{2}$. (3)

UNIT II

13. (a) If $u = \log r$ where $r^2 = (x-\alpha)^2 + (y-\alpha)^2$ and $(x-\alpha)$ and $(y-b)$ are not simultaneous zero, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. (6)

(b) Expand $x^3y + 3y - 2$ in power of $(x-1)$ and $(y+2)$ upto the third degree terms. (5)

Or

14. (a) If $x = e^r \sec\theta$, $y = e^r \tan\theta$ prove $\frac{\partial(x,\ y)}{\partial(r,\ \theta)} \frac{\partial(r,\ \theta)}{\partial(x,\ y)} = 1$ (5)

(b) Find the volume of the greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. (6)

UNIT III

15 (a) Evaluate $\int\limits_0^3 \int\limits_1^{\sqrt{4-y}} dx\ dy$ by changing the order of integration.

(b) Change the order of integration in $I = \int\limits_{-a}^{a} \int\limits_0^{\sqrt{a^2-x^2}}$ $(x^2+y^2)dx\ dy$.

Or

16. Find the volume of that portion of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} = 1$ which lies in the first using triple integration.

UNIT IV

17. (a) Find the orthogonal trajectories on the conicoid $(x+y)z = 0$ of the conics in which it is cut by the system of planes $x - y + z = k$, where k is a parameter. (6)

(b) Solve $(px - y)(py + x) = 2p$. (5)

Or

18. (a) Solve $p^2 + 2yp \cot x - y^2 = 0$. (5)

(b) Solve $y = 3px + 6p^2y^2$. (6)

UNIT V

19. (a) Solve $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$. (5)

(b) Solve $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = \sin(\log x)$. (6)

Or

20. Solve the simultaneous equations and $\frac{d^2y}{dt^2} + y = te^{3t}$ and $\frac{d^2y}{dt^2} + y - 2x = \cos^2 t$.

B.Tech. DEGREE EXAMINATION,
NOVEMBER 2016.
First Semester
MATHEMATICS – I
(Common to All branches)
(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Write down the formula for radius of curvature in Cartesian coordinates.
2. Write down the relationship between Beta and Gamma functions.
3. If $u = (x-y)^4 + (y-z)^4 + (z-x)^4$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
4. In polar coordinates $x = r\cos\theta, y = r\sin\theta$, show that $\frac{\partial(x,y)}{\partial(r,\theta)} = r$.
5. Evaluate $\int_0^a \int_0^b xy(x-y)dxdy$.
6. Write down the total volume of the solid of revolution.
7. Solve: $ydx - xdy - 3x^2y^2e^{x3}dx = 0$.
8. Solve: $p^2 - 9p + 18 = 0$ where $p = \frac{dy}{dx}$.
9. Solve: $(D^3 + D^2 - D - 1)y = 0$.

10. Find the P.I of the differential equation $(D^2 - 2D + 1)y = e^x \sin x$.

PART B – $(5 \times 11 = 55$ marks)

Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. (a) Find the radius of curvature at the point $(\alpha \cos^3 \theta, \alpha \sin^3 \theta)$ on the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = \alpha^{\frac{2}{3}}$. (6)

(b) Evaluate $\int_0^{\frac{\pi}{2}} \cos^9 \theta d\theta$ using gamma function. (5)

Or

12. Show that the center of curvature at any point (x, y) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(\frac{a^2 - b^2}{a^4} x^3, \frac{a^2 - b^2}{-b^4} y^3\right)$ and hence show that the equation of the evolute is $(ax)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}}$. (11)

UNIT II

13. (a) If $u = \sin^{-1}\left(\frac{x + y}{\sqrt{x} + \sqrt{y}}\right)$ then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2} \tan u$. (5)

(b) If $x = r\cos\theta, y = r\sin\theta$ find $\frac{\partial(x, y)}{\partial(r, \theta)}$ and $\frac{\partial(r, \theta)}{\partial(x, y)}$ then verify that $\frac{\partial(x, y)}{\partial(r, \theta)} \frac{\partial(r, \theta)}{\partial(x, y)} = 1$. (6)

Or

14. Expand $xy^2 + 2x - 3y$ in powers of $(x + 2)$ and $(y - 1)$ upto the degree terms. (11)

UNIT III

15 Evaluate by changing to polars, the integral $\int_0^a \int_y^b \frac{x^2 dxdy}{\sqrt{x^2+y^2}}$. (11)

Or

16. Find the volume of that portion of the elipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} = 1$, which lies in the first octant. (11)

UNIT IV

17. (a) Solve: $(x^4e^x - 2mxy)dx + 2mx^2ydy = 0$. (5)

    (b) Solve $(x^2 - yx^2)\frac{dy}{dx} + (y^2 + x^2y^2) = 0$. (6)

Or

18. Solve: $p^3 - 4xyp + 8y^2 = 0$. (11)

UNIT V

19. Solve the simultaneous equations $\frac{dx}{dt} + 2x - 3y = t; \frac{dy}{dt} - 3x + 2y = e^{2t}$. (11)

Or

20. Solve by the method of variation of parameters, the equations $\frac{d^2y}{dx^2} + 4y = 4\tan 2x$. (11)

B.Tech. DEGREE EXAMINATION, MAY 2016.
First Semester
MATHEMATICS – I
(Common to All branches)
(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A – (10 × 2 = 20 marks)
Answer ALL Questions
All questions carry equal marks

1. Find the radius of curvature of the curve $y = e^x$ at $(0, 1)$.

2. Define curvature of a curve.

3. Find $\dfrac{dy}{dx}$ when $x^3 + y^3 = 3axy$.

4. If $x = u(1 + v)$ and $y = v(1 + u)$ find $\dfrac{\partial(x, y)}{\partial(u, v)}$

5. Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{\sin\theta} r d\theta dr$.

6. Evaluate $\int_0^a \int_0^b \int_0^c (x^2 + y^2 + z^2) dxdydz$.

7. Solve: $p^2 - 7p + 12 = 0$.

8. Solve: $(y - px)(p - 1) = p$.

9. Solve: $(D^3 - D^2 - D + 1)y = 0$.

10. Solve $y' + y = \sin x$ using the method of variation of parameters.

PART B − (5 × 11 = 55 marks)
Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. Find the radius of curvature at the point $(\alpha\cos^3\theta, \alpha\sin^3\theta)$ on the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = \alpha^{\frac{2}{3}}$. Also show that $\rho^3 = 27ax$.

Or

12. Find the evolute of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.

UNIT II

13. If $x = u(1-v), y = uv$ then compute J and J' and prove that $JJ' = 1$.

Or

14. Find the greatest and least distance of the point (3, 4, 12) from the unit sphere whose centre is at the orgin.

UNIT III

15 Change the order of integration and hence evaluate $\int\limits_0^{12}\int\limits_{x^2}^{-x} xydy\,dx$.

Or

16. Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ without transformation.

UNIT IV

17. Solve $p^2 + 2py\cot x = y^2$.

Or

18. Solve: $y(2x^2y + e^x)dx - (e^x + y^3)dy = 0$.

UNIT V

19. Solve $(D^2 - 4D + 4)y = \cos h2x + 3$.

Or

20. Solve $\dfrac{dx}{dt} - y = t; \dfrac{dy}{dt} + x = 1$.

B.Tech. DEGREE EXAMINATION, JANUARY 2016.
First Semester
MATHEMATICS – I
(Common to All branches)
(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Find the radius of curvature of the curve $y = 2x^3 - 3x + 2$ at $(1, 1)$.
2. Define evolute and involute.
3. Find the stationary points of $(x, y) = x^2 - xy + y^2 - 2x + y$.
4. If $x = r\cos\theta, y = r\sin\theta$ find $\dfrac{(x, y)}{(r, \theta)}$.
5. Evaluate $\int\limits_2^b \int\limits_2^a \dfrac{dx\ dy}{xy}$.
6. Evaluate $\int\limits_0^\infty \int\limits_x^\infty \dfrac{e^{-y}}{y} dy\ dx$ by changing the order of integration.
7. Solve: $(y - px)(p - 1) = p$.
8. Solve: $p^2 - 7p + 12 = 0$.
9. Solve: $(D^2 + 3D + 2)y = 0$.
10. Eliminate y from the simultaneous equations $\dfrac{dx}{dt} + y = \sin t$; $\dfrac{dy}{dt} + x = \cos t$.

PART B – (5 × 11 = 55 marks)
Answer ALL the Units, choosing ONE full question
from each unit

UNIT I

11. Find the radius of curvature at the $(\alpha, 0)$ on the curve $xy^2 = \alpha^3 - x^3$.

Or

12. Prove that the evolute of the rectangular hyperbola $xy = c^2$ is $(x+y)^{2/3} - (x-y)^{2/3} = (4c)^{2/3}$.

UNIT II

13. Find the Taylor's series expansion of $f(x, y) = x^2y + \sin y + e^x$ about the point $(1, \pi)$.

Or

14. Find the maximum and minimum values of $f(x, y) = 2(x^2 - y^2) - x^4 + y^4$.

UNIT III

15 Evaluate $\int\limits_0^1 \int\limits_{x^2}^{2-x} xy\, dy\, dx$ by changing the order of integration.

Or

16. Evaluate $\int\limits_0^1 \int\limits_0^{1-x} \int\limits_0^{1-x-y} xyz dx\ dy\ dz$.

UNIT IV

17. Solve $p^2 + 2py \cot x = y^2$.

Or

18. Solve: $y = 2px + p^2$.

UNIT V

19. Solve $(D^2 - 3D + 2)y = 2\cos(2x + 3) + 2e^x$.

Or

20. Solve $(D^2 + a^2)y = \tan ax$ by the method of variation of parameters.

B.Tech. DEGREE EXAMINATION, JANUARY 2015.
First Semester
(Common to All branches)
MATHEMATICS – I

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Define curvature and radius of curvature.
2. What is the relation between gamma and beta function.
3. If $u = \frac{x}{y} + \frac{y}{z}$ find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$
4. If $x = r\cos\theta, y = r\sin\theta$ find $\frac{\partial(x,y)}{\partial(r,\theta)}$.
5. Evaluate $\int_0^1 \int_1^2 x(x+y)dydx$.
6. Change the order of the integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}dydx$.
7. Solve: $(y\cos x + 1)dx + \sin x dy = 0$.
8. Solve: $(y - px)(p-1) = p$.
9. Solve: $(D^3 - 12D + 16)y = 0$.
10. Find the particular integral of $(D^2+1)y = \sin x$.

PART B – $(5 \times 11 = 55$ marks$)$
Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. (a) Prove that the radius of curvature at any point of the cycloid. $x = a(\theta + \sin\theta), y = \alpha(1 - \cos\theta)$ is $4\alpha \cos \frac{\theta}{2}$. (6) (b) Find the radius of curvature at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on the curve $x^3 + y^3 + 3axy = 0$. (5)

Or

12. Find the equation of the evolute of he parabola $y^2 = 4ax$.

UNIT II

13. (a) if z is a function of x and y and u and v are other two variable, such that $u = lx + my, v = ly - mx$. Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$. (6)
(b) Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.

Or

14. (a) Expand $e^x \log(1 + y)$ in powers of x and y up to terms of third degree. (5)
(b) A rectangular box open at the top, is to have a volume of 32cc. Find the dimensions of the box, the requires the least material for its construction. (6)

UNIT III

15 Change the order of integration and hence evaluate $\int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx$. (11)

Or

16. Find the volume of that portion of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ which lies in the first octant using triple integration. (11)

UNIT IV

17. Solve $p^2 + 2py\cot x = y^2$.

Or

18. Solve: $x(1-x^2)\frac{dy}{dx} + (2x^2-1)y = x^3$.

UNIT V

19. Solve the method of variation of parameter $\frac{d^2y}{dx^2} + y = \sin x$. (11)

Or

20. Solve $\frac{dx}{dt} + y = \sin t, \frac{dy}{dt} + x = \cos t$ given $x = 2, y = 0$ at $t = 0$. (11)

B.Tech. DEGREE EXAMINATION, JANUARY 2014.
First Semester
MATHEMATICS – I
(Common to All branches)

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Define curvature of a straight line $y = 2x + 3$ at any of its point.
2. Write the relationship between Beta and Gamma function.
3. State chain rule for total derivative of $u(x, y)$ where $x = f(t)$ and $y = g(t)$.
4. Expand $e^x \log(1+y)$ in powers of x and y upto terms of second degree.
5. Evaluate $\int\limits_1^2 \int\limits_2^5 xy\ dx\ dy$.
6. Find the volume of a cube of side of length two.
7. State the necessary and sufficient condition for the differential equation $M(x, y)dx + N(x, y)dy = 0$ to be exact.
8. Solve: $p = sin(y - px)$.
9. Find the particular integral of $\frac{d^2y}{dx^2} + y = x$.
10. Solve: $x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$.

PART B – ($5 \times 11 = 55$ marks)
Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. (a) Find the circle of curvature at the point $\left(\frac{1}{4}, \frac{1}{4}\right)$ for the curve $\sqrt{x} + \sqrt{y} = 1$. (5) (b) Show that $\beta(m, n) = \frac{\Gamma_m \Gamma_n}{\Gamma_{m+n}}$ and hence deduce $\Gamma_{\frac{1}{2}} = \sqrt{\pi}$. (6)

Or

12. (a) Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (5) (b) Show that $\beta(m, \frac{1}{2}) = 2^{2m-1}\beta(m, n)$. (6)

UNIT II

13. (a) If $u = x^2 + y^2 + z^2, v = xy + yz + zx, w = x + y + z$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ (6)
(b) Find the maximum and minimum of $f(x, y) = x^3 + y^3 - 3xy$.

Or

14. (a) If $u = f(x - y, y - z, z - x)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (5)
(b) Find the minimum value of $x^2+y^2+z^2$ where $ax+by+cz = p$ (6)

UNIT III

15 Show that area between two parabolas $y^2 = 4x$ and $x^2 = 4y$ is $\frac{16}{3}$ by double integrals. (5)

Or

16. (a) Change the order of integration $\int_0^1 \int_y^1 \frac{xdxdy}{x^2 + y^2}$. (5)

(b) Find the volume of the sphere $x^2 + y^2 + z^2 = 1$ by triple integrals. (6)

UNIT IV

17. (a) Solve: $\frac{dy}{dx} = y \tan x - y^2 \sec x$. (5)

(b) Find the orthogonal trajectories of the family of the parabolas $y^2 = 4ax$. (6)

Or

18. (a) Solve: $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$. (5)

(b) Solve: $y = 2px + y^2p^3$. (6)

UNIT V

19. (a) Solve: $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = \log x$. (6)

(b) Solve by the of variation of parameters $\frac{d^2y}{dx^2} + y = \sec x$. (5)

Or

20. Solve: $\frac{dx}{dt} + y = \sin t$.

$\frac{dy}{dt} + x = \cos t$ (11)

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014.
First/Second Semester
(Common to All branches)
MATHEMATICS – I

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)
Answer ALL Questions
All questions carry equal marks

1. Write down the formula for radius of curvature in terms of parametric co-ordinates

2. Write the relationship between gamma and beta functions.

3. Find the slope of tangent at the point (x_1, y_1) of the curve $3x^2 + 2xy - y^3 + 3 = 0$.

4. State any two properties of Jacobians.

5. Define area in polar co-ordinates.

6. Evaluate $\int\limits_0^a \int\limits_0^b \int\limits_0^c xyzdzdydx$.

7. Write the general solution of $= px + (a/p)$.

8. Solve: $P^2 - 7p + 12 = 0$.

9. Find the particular integral of the differential equation $(D^2 - 4D + 13)y = e^{2x}$.

10. Find the complementary function of $(D^2 - 2D + 1)y = x$.

PART B – ($5 \times 11 = 55$ marks)
Answer ALL the Units, choosing ONE full question from each unit

UNIT I

11. (a) Find radius of curvature at t on $x = 6t^2 - 3t^4, y = 8t^3$. (6) (b) Find the equation of circle of curvature at (c, c) on $xy = c^2$. (5)

Or

12. Find the equation of the evolute of the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ (11)

UNIT II

13. (a) If $y_1 = \dfrac{x_2x_3}{x_1}, y_2 = \dfrac{x_3x_1}{x_2}, y_3 = \dfrac{x_1x_2}{x_3}$ show that $\dfrac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = 4$. (6)
(b) Expand $e^x \cos y$ at $(0, 0)$ upto third degree using Taylor's series. (5)

Or

14. A rectangular box open at the top is to have a volume of 32 c.c. Find the dimensions of the box that requires the least materials for its construction. (11)

UNIT III

15 (a) Change the order of integration and hence find the value of $\int\limits_0^a \int\limits_a^x x^2 + y^2 dy\ dx$. (6) (b) Find the area enclosed by the lines $x = 0, x = 2, y = 0, y = 2$. (5)

Or

16. Find the volume bounded by cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$. (11)

UNIT IV

17. (a) Show that $(5x^4+3x^2y^2-2xy^3)dx+(2x^3y-3x^2y^2-5y^4)dy = 0$ is an exact equation and hence solve it. (6)
(b) Find the orthogonal trajectories of the system of parabolas $y^2 = 4zx$. (5)

Or

18. (a) Solve: $P^3 - 4xyp + 8y^2$. (6)
(b) Solve: $y = (x - a)p - p^2$ using Clairaut's type. (5)

UNIT V

19. (a) Solve: $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$. (6)
(b) Solve by the of variation of parameters $\dfrac{d^2y}{dx^2}+y = \sec x$. (5)

Or

20. Solve: $\dfrac{dx}{dt} + 2x - 3y = t, \dfrac{dy}{dt} - 3x + 2y = e^2t$. (11)